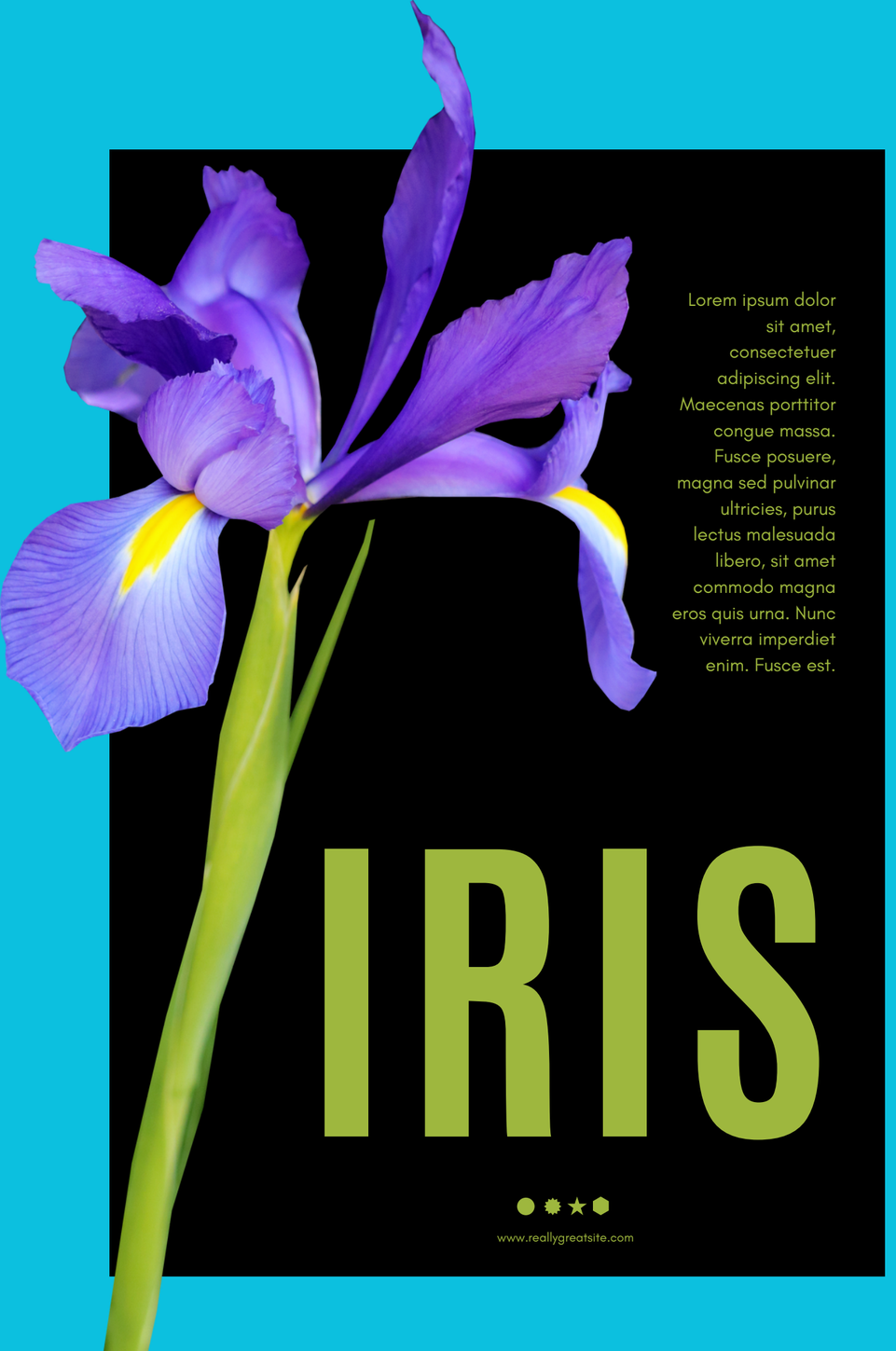
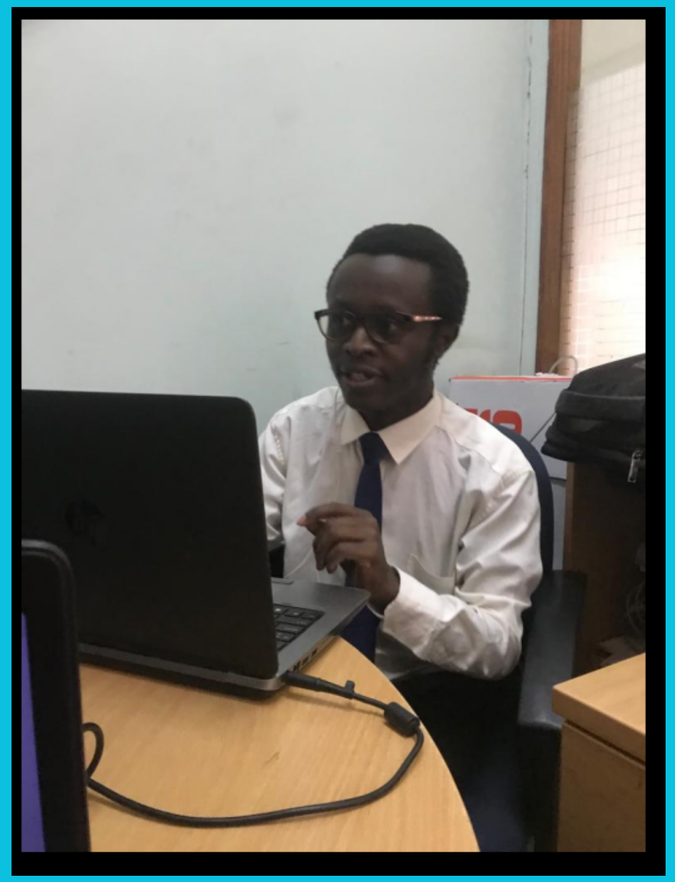


Iris Flower Classification Project Presentation

Data Scientist and Machine Learning Engineer



NICHOLAS MUCHIRI KINYUA

https://github.com/MuchiriKinyua

muchiri.kinyua6564@gmail.com

“ Data is the new oil ~Muchiri Kinyua

Content Outline

Topics for discussion

01 Business Understanding

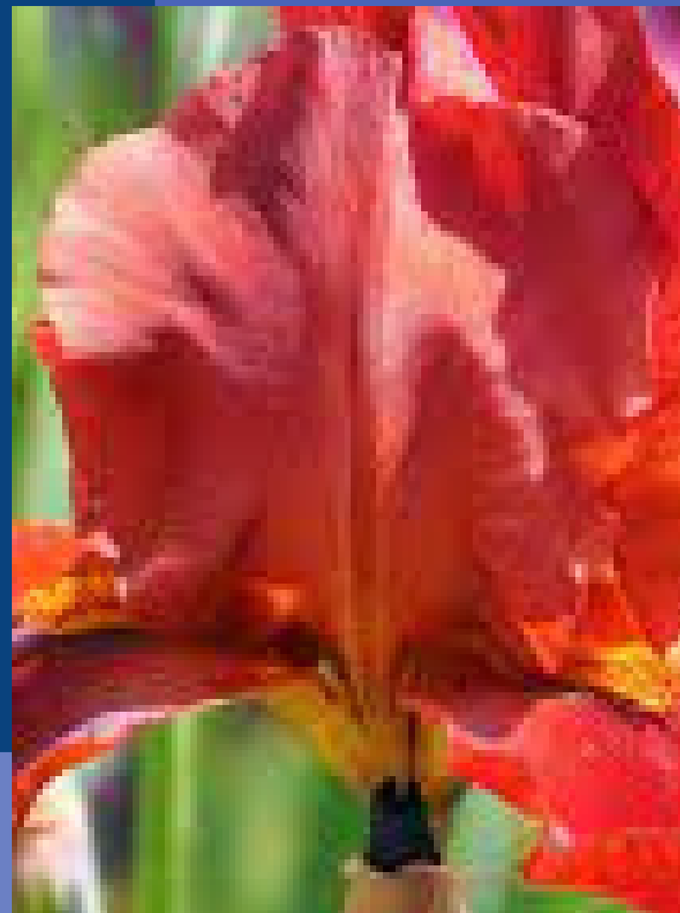
02 Problem Statement

03 Data modelling

04 Model interpretation

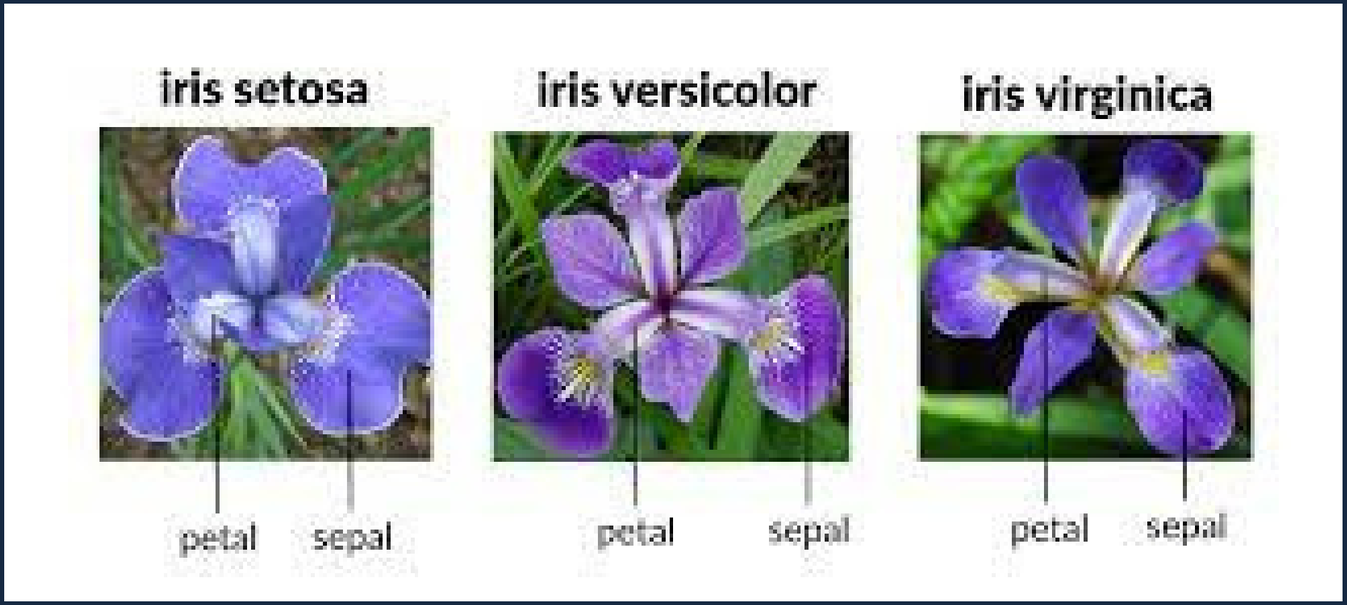
05 Conclusions

06 Recommendations



Business Understanding

This project is aimed to classify iris flowers into three species (setosa, versicolor, virginica) based on sepal length, sepal width, petal length, and petal width.



PROBLEM STATEMENT

The problem at hand involves the classification of iris p
lant species based on their distinct characteristics, specifically the measurements of sepal length, sepal width, petal length, and petal width. With the availability of the Iris dataset, which comprises data samples of various iris plants labeled with their corresponding species (setosa, versicolor, or virginica).

This classification task is vital for botanists, researchers, and horticulturists seeking to understand and differentiate between different iris species for various purposes, including plant taxonomy, ecological studies, and cultivation practices. By leveraging machine learning techniques, we aim to create models that can effectively learn and generalize patterns from the provided dataset to accurately classify iris plants into their respective species based on their morphological attributes.

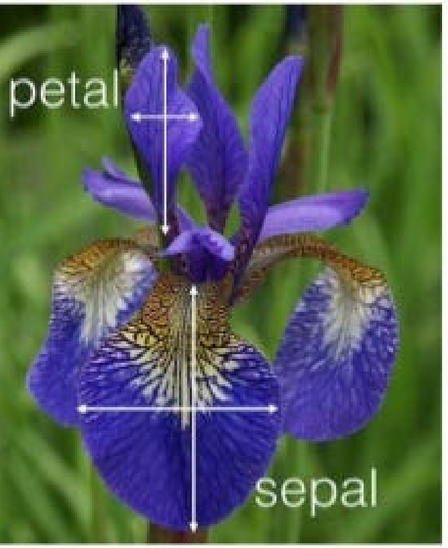
MAIN OBJECTIVE

The objective is to develop accurate machine learning models capable of classifying new iris plants into their correct species categories.

Specific Objectives

To build a model that can accurately classify Iris flowers into their three known species (Iris Setosa, Iris Versicolor, Iris Virginica) based on the provided measurements.

Supervised learning *classification* problem
(using the Iris flower data set)



Training / test data

Features				Labels
Sepal length	Sepal width	Petal length	Petal width	Species
5.1	3.5	1.4	0.2	Iris setosa
4.9	3.0	1.4	0.2	Iris setosa
7.0	3.2	4.7	1.4	Iris versicolor
6.4	3.2	4.5	1.5	Iris versicolor
6.3	3.3	6.0	2.5	Iris virginica
5.8	3.3	6.0	2.5	Iris virginica

To compare the classification accuracy achieved using PCA. This will help in understanding if dimensionality reduction through PCA benefits the model performance in this specific case.

To evaluate the performance of different models using appropriate metrics such as accuracy

DATA UNDERSTANDING

The data includes three iris species with 50 samples each as well as some properties about each flower. One flower species is linearly separable from the other two, but the other two are not linearly separable from each other. The dataset, Iris, has got 150 rows and 6 columns. columns ae as follows:

Id - The primary identifier

sepal length (cm) - length of the sepal in centimeters

sepal width (cm) - width of the sepal in centimeters

petal length (cm) - length of the petal in centimeters

petal width (cm) - width of the petal in centimeters

Target - Setosa, Versicolor, Virginica

DATA ANALYSIS

97% of the data collected was used.

DATA ANALYSIS ON IRIS DATA SET

MODELS USED

Model 1:
Logistic regression

The model appears to have an excellent performance, with an accuracy of 95%

Model 2:
Decision Trees

Decision Tree model achieved an accuracy of 94%.

Model 3:
Random Forest

Random Forest achieved an accuracy of 94%.

Model 4:
Neural Networks

Neural Networks chieved an accuracy of 94%.

Model 5:
KNN

KNN achieved an accuracy of 94%.

Model 6:
Nayes Bayes

Naive Bayes achieved an accuracy of 94%.

Evaluation

Evaluation
Ease of Interpretation: Accuracy is an intuitive and easily interpretable metric. It represents the percentage of correct predictions among all predictions made by the model. Users, including hotel staff, can readily understand and trust this metric.

Clear Benchmark: Accuracy provides a clear benchmark for evaluating model performance. It answers the basic question: "How often is the model correct?" This simplicity can be advantageous for communication and decision-making.

Balanced Classes: If the classes of interest (e.g., setosa, versicolor, virginica) are roughly balanced, accuracy can be an effective measure. It doesn't favor one class over another and provides a sense of overall correctness.

I tested and from that we choose the best 3 performing models which are :

Logistic regression = 95%
Random Forest = 94%
KNN = 94%

Conclusions and Recommendations

1. It provides a good example for exploring various machine learning techniques, particularly classification algorithms.
2. Petal length and sepal length are twice the size of petal width and sepal width.
3. Feature importance analysis: It can provide insights into which features contribute the most to the classification task. This information can be valuable for understanding the underlying characteristics of the data and potentially simplifying the model by focusing on the most relevant features.
4. Outlier detection and removal, as demonstrated in the analysis, can improve model performance by reducing the impact of noisy data points. However, it's essential to exercise caution and consider the domain knowledge when deciding how to handle outliers like I did. But mostly justify.
5. It's essential to monitor the model's performance over time and periodically retrain it with new data to ensure its effectiveness.

Questions or comments?

Get in touch!

Let's talk

Instructions: Feel free to ask questions at any time during or after the presentation. I'll address them shortly.

Contact Information: For any follow-up questions or discussions, please reach out to muchiri.kinyua6564@gmail.com.

You can also find the full project at: https://github.com/MuchiriKinyua/iris-project/tree/main

Thank you